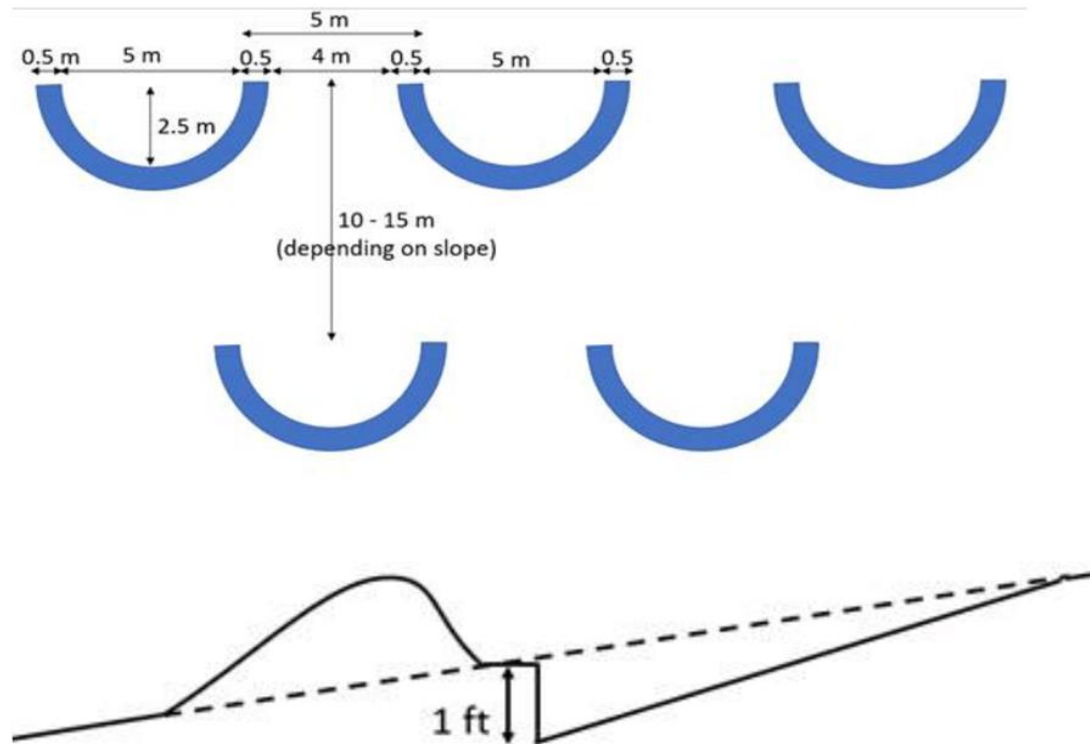


Digging Bunds

Description

A bund is a half-moon-shaped basin dug in the earth perpendicular to the slope of the terrain to retain rain, improve soil moisture, and reduce soil erosion. The technique of creating Bunds belongs to the overall category of water harvesting interventions. This type of micro catchment water harvesting technique is suitable for slopes up to 15%, however bunds made of earth are seldom used in areas with slopes greater than 5% with a precipitation rate higher than 300 mm/y. At lesser slopes, Demi-lunes are also used in areas with higher rain falls.



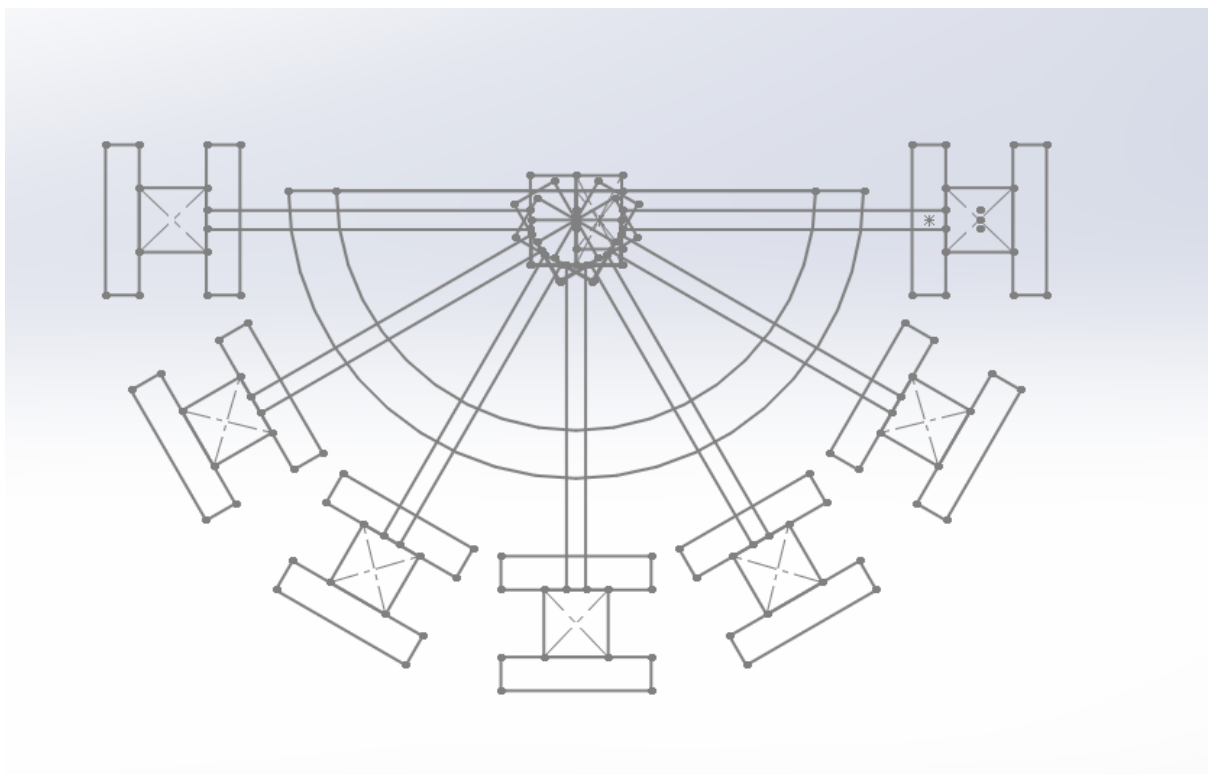
Bund Specification

A bund is a half-moon-shaped basin with a rounded embankment (ridge). The diameter of a bund is variable depending on the terrain. But usually, it is around 5 meters. The basin is around 30 cm deep. The distance between two bunds is normally 4 meters and the next row of bunds down the slope is between 10 and 15 meters depending on the slope (see figure).

Excavating Bunds

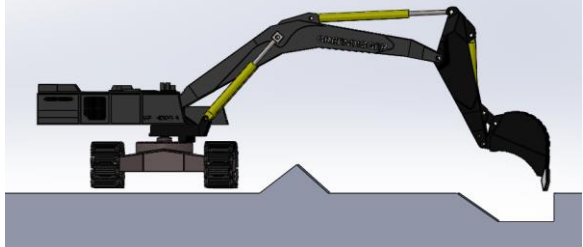
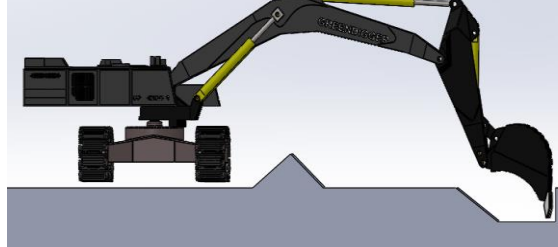
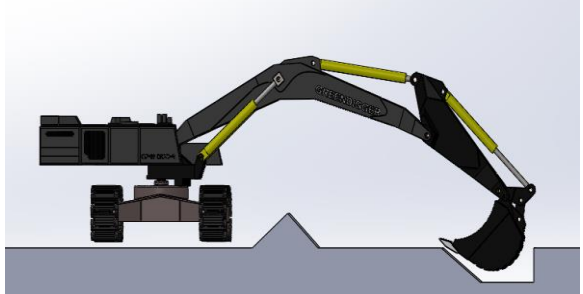
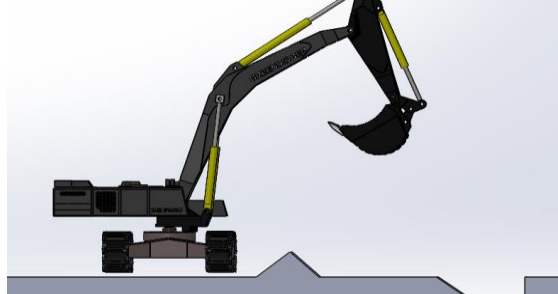
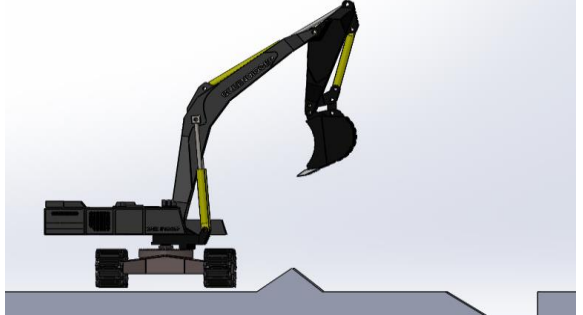
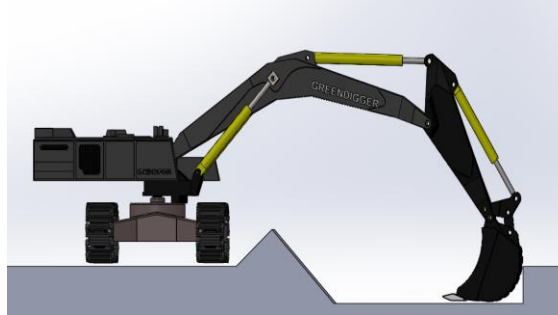
Please note that bunds placement and dimensions are not crucial and may vary. On the other hand, the ridge should be continuous.

First, determine the slope angle, ensuring that the bund is built perpendicular to the slope. Scan for any objects, the bund, and the workspace. Determine the location and the depth for digging and decide on the machine's location.

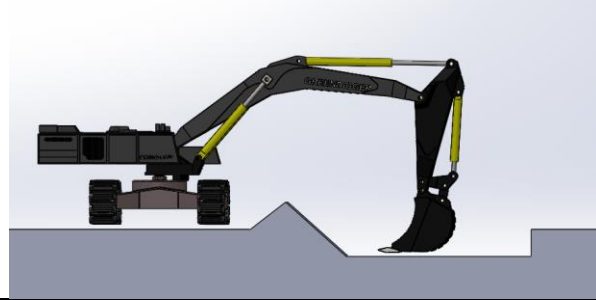


Excavator Movement

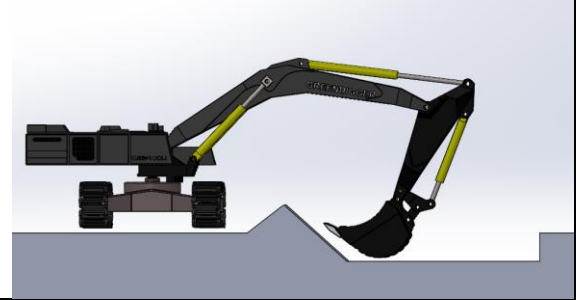
The GreenDigger starts on the outer edge of the bund. The excavator arm scoops from the center of the bund towards the GreenDigger. Scraping the soil towards the machine, creating a basin in the middle and a ridge towards the machine. Beginning from the outer side of the bund the GreenDigger moves in a half-circle around the to be constructed Bund (see figure).

Step 1: GreenDigger stands at the edge of the bund and places the digging bucket above the centre point of the bund.	Step 2: The teeth of the digging bucket stick perpendicularly into the ground. Let the teeth and shovel do the work.
	
Step 3: With a smooth scooping motion, the first earth is scooped out	Step 4: The earth is unloaded in front of the machine; this is the beginning of the ridge.
	
Step 5: This process can be repeated if necessary.	Step 6: Now we use the teeth and shovel to scoop up the remaining earth
	

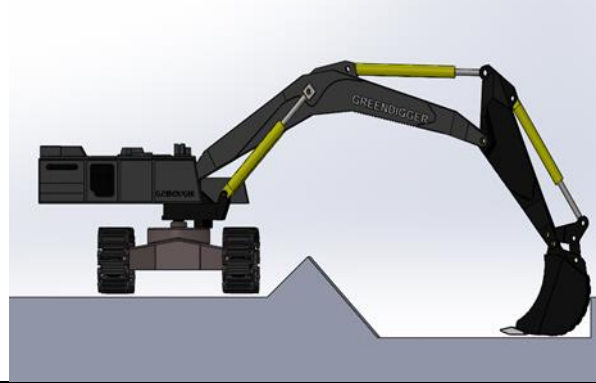
Step 7: Finally, we can scrape over the soil several more times.



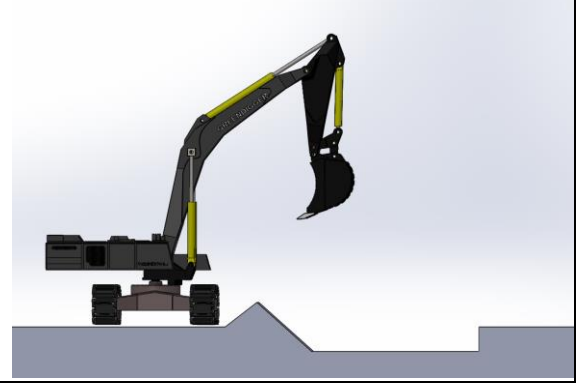
Step 8: This creates a solid dam.



Step 9: If necessary, repeat the scraping or shovelling.



Step 10: Move to the next digging position and repeat the process.



Field test results

Digging bunds requires a fairly standard excavation movement. The GreenDigger can move slowly, speed is not a necessity. The GreenDigger does not need accuracy (except building the ridge) but can work coarsely. The environmental safety settings for obstacles and people/animals can be set broadly.

The main problem occurs when the digging bucket gets stuck on an underground rock, or other obstacle not spotted in the survey. This will cause one of the actuators (or cylinders) to experience too much resistance and in the end titling of the GreenDigger. At this moment the GreenDigger will stop excavating.

Depending on the severity of the problem restart options must be designed.